

AdaptiveFly: Bio-Congruent Adaptation for Embodied 3D Locomotion

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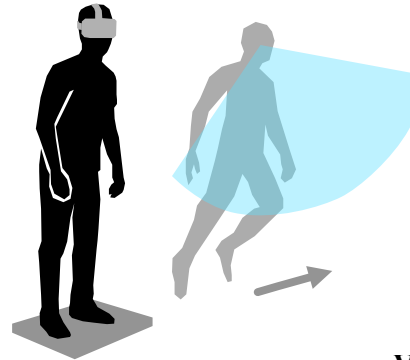
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(a) User Experience



(b) Exploration/Observation



(c) Fast Steering/ Turning

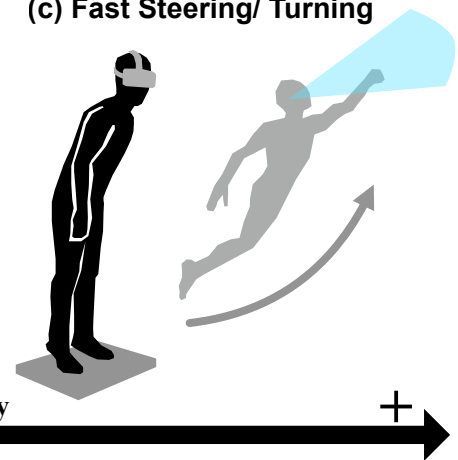


Figure 1: Concept of the proposed interaction technique. (a) User experience: The user stands on a sensor-equipped platform and wears a VR headset to engage with the system. (b) Exploration / Observation mode (low speed): When the user leans forward slightly, the system detects a low velocity and expands the head-control deadzone, allowing free head rotation for environmental observation. In this state, gaze direction is decoupled from movement direction. (c) Fast Steering / Turning mode (high speed): When the user leans forward significantly, the velocity increases and the deadzone shrinks. In this state, head rotation directly affects the flight direction, enabling fast and agile maneuvering.

Abstract

Virtual reality (VR) offers compelling opportunities for human augmentation, with free three-dimensional (3D) flight enabling fluid navigation and immersive spatial exploration. However, most

existing VR flight techniques rely either on abstract input methods—leading to sensorimotor mismatch and limited natural interaction—or on static embodied mappings that cannot adapt to the dynamic nature of movement. We introduce AdaptiveFly, an embodied flight technique based on bio-congruent adaptation. AdaptiveFly dynamically modulates the coupling between head rotation and yaw control according to user speed—loosening it at low speeds to allow free gaze movement, and tightening it at high speeds for stable steering. It also calibrates control sensitivity to individual posture characteristics and incorporates asymmetric pitch mapping to enhance the intuitiveness of vertical motion. A user study



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confirms that AdaptiveFly enhances naturalness, comfort, and coordination compared to both abstract and static embodied baselines. A proof-of-concept deployment on a physical drone further demonstrates the technique’s real-world applicability. This study expands the design of embodied VR locomotion and demonstrates the potential value of adaptive, human-consistent control strategies in 3D interfaces and human augmentation technologies.

CCS Concepts

• **Human-centered computing** → **Virtual reality**; **Interaction paradigms**; *Empirical studies in interaction design*.

Keywords

Virtual Reality, Locomotion Techniques, Embodied Interaction, Drone Teleoperation, Human-Robot Interaction

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1 Introduction

Virtual reality (VR) provides new opportunities for extending human capabilities beyond physical limitations, enabling experiences that are unattainable in the physical world. Among these, free three-dimensional (3D) flight stands out as a notable form of human augmentation. Flying allows users to move fluidly through space, observe environments from novel vantage points, and traverse large or multi-layered structures with ease [24, 30]. Such capabilities are valuable across domains including spatial exploration, scientific visualization, environmental education, and creative design. In these scenarios, users must often navigate while continuously inspecting their surroundings, making it essential to support movement and free visual exploration simultaneously.

Most existing VR systems implement 3D movement and flight using abstract input mappings such as joysticks, buttons, or thumbsticks. Although easy to implement and broadly compatible, these mappings lack body-related motion cues and do not align with users’ natural sensorimotor coordination [8]. As a result, they frequently induce visual-vestibular conflict, reduce immersion, and constrain the user’s ability to observe and navigate simultaneously. Furthermore, abstract inputs typically occupy the user’s hands, limiting opportunities for interaction during flight-based tasks that require both mobility and environmental awareness.

To improve naturalness and enhance perceptual freedom, researchers have proposed embodied flight interfaces that rely on body leaning or posture adjustments [26, 27]. These techniques introduce body-related cues and can reduce discomfort relative to abstract mappings [1]. However, most existing embodied flight systems depend on static input-output relationships that fail to accommodate the dynamic requirements of 3D locomotion. During natural locomotion, users shift between low-speed exploration—requiring broad, unconstrained head movements—and high-speed travel,

where tighter directional coupling becomes necessary. Static mappings cannot reconcile these opposing demands, often leading to unintended activations, inconsistent directional control, or reduced transparency.

This limitation points to a broader gap in current locomotion research: effective embodied 3D locomotion requires control strategies that adapt in ways consistent with human sensorimotor behavior. We refer to this as bio-congruent adaptation for embodied 3D locomotion. Such adaptation aligns system behavior with the natural, context-dependent adjustments humans make when coordinating gaze, head, and body during movement.

To implement this principle, we present AdaptiveFly, a 4-degree-of-freedom embodied flight technique grounded in bio-congruent adaptive mechanisms. AdaptiveFly dynamically adjusts the coupling between head rotation and yaw control based on the user’s current speed: at low speeds, the system weakens this coupling to support free and uninterrupted visual exploration; at high speeds, it strengthens the coupling to maintain intuitive and responsive steering. Additionally, the system automatically calibrates control sensitivity to the user’s individual posture characteristics, reducing unintentional input. An asymmetric pitch mapping founded on human gaze biomechanics further enhances comfort and intuitiveness during vertical movements. Together, these components instantiate bio-congruent adaptation within a practical 3D locomotion technique, enabling users to explore virtual environments freely while maintaining a fluid and intuitive sense of movement. Our contributions are as follows:

- We introduce AdaptiveFly, an embodied, context-adaptive 3D flight technique that applies the concept of bio-congruent adaptation for embodied 3D locomotion and supports free visual exploration during movement.
- Through empirical evaluation, we demonstrate its experiential benefits over abstract and static embodied baselines in terms of naturalness, comfort, and coordination between looking and steering.
- Based on our findings, we provide design implications for human-augmentation and 3D user interface research, highlighting the importance of context-adaptive and bio-congruent control in achieving natural embodied locomotion.

2 Related Works

2.1 3D Locomotion in Virtual Environments

Locomotion techniques in virtual environments have seen diverse technical approaches proposed to address the requirement of exploring vast virtual spaces within finite physical spaces. Techniques such as redirected walking [22] and omnidirectional treadmills [16] enable locomotion across vast VR spaces within small physical spaces by altering the correspondence between the user’s walking actions and actual physical displacement. Additionally, methods applying motion-based walking metaphors that can be performed while remaining stationary, such as Walking-in-Place (WIP) [3] and arm-swinging techniques [25], have also been proposed. These methods are all characterized by their ability to reference the ecological standard of real-world walking. In contrast, VR affords the possibility of designing three-dimensional locomotion techniques

that would be impossible in the physical world. However, the lack of innate mental models for such modalities complicates the design strategy of natural interfaces [29]. Some 3D locomotion methods intentionally embrace physically impossible or non-ecological interaction metaphors. Techniques such as Teleportation enhance task efficiency and usability but disrupt spatial continuity, often resulting in disorientation [6]. World-in-Miniature (WIM) allows users to traverse vast distances instantly or manipulate their position from a god-like perspective, but it introduces an exocentric viewpoint that can break embodied continuity and requires additional cognitive effort to reconcile multiple scales of representation [34].

Parallel to these developments, many locomotion techniques can be organized along a spectrum of motor input abstraction. At one end of this spectrum are abstract controller-based interfaces that treat physical motions as symbolic commands. Joystick-based locomotion, for example, offers precision and low fatigue but tends to induce cybersickness due to visual-vestibular conflict [8]. At the opposite end are physically based mappings, in which metaphorical interpretations of bodily movements are directly coupled to locomotion behaviors. Examples include bird-like flapping gestures [24] or grab-and-pull interactions [30]. While such embodied techniques are reported to enhance the sense of agency and immersion, they are associated with difficulties in precise manipulation and significant physical load and fatigue [1]. Between these extremes lie hybrid techniques that blend physical metaphors with abstract control properties. For example, Leaning techniques that map trunk inclination to propulsion [26, 27] and Gaze-directed techniques that steer movement in the direction of gaze [5] retain physical movement metaphors while possessing control characteristics similar to abstract inputs such as joysticks. Multiple studies suggest that these methods may contribute to mitigating cybersickness due to the relatively high congruence between visual motion and postural changes [1, 26]. Although no universal “gold standard” exists for 3D locomotion, research is progressing toward identifying an optimal balance along the spectrum—leveraging appropriately abstracted physical movements to achieve comfortable, immersive, and intuitively controllable 3D locomotion.

2.2 Coupling of Body Motion in Motion-based Locomotion

While locomotion techniques that incorporate physical movement generally offer greater intuitiveness and immersion than abstract controller-based inputs, they face a structural limitation: the natural movement of body parts becomes intrinsically and continuously coupled with the system’s locomotion control. This tight coupling risks misinterpreting exploratory or task-related movements as locomotion commands.

Gaze-oriented locomotion, which uses HMD orientation or eye gaze as the travel vector, has become a default choice in many VR applications due to ecological evidence that head and gaze typically precede the direction of travel [2, 11]. However, because this method algorithmically equates gaze direction with intended movement direction, it inhibits the natural behavior of “looking around while walking forward,” producing what is known as the Look–Steer Conflict [4]. This issue has the same root as the Midas Touch problem,

where “looking constitutes selection” [17], and results from the system’s over-interpretation of gaze as an indicator of locomotor intent. A conventional solution is to introduce a dead zone—an interval in which the system does not respond until gaze deviation exceeds a predefined threshold. Nonlinear control schemes have also been shown to alleviate this issue [9]. Other countermeasures include suppressing unintended actuation by gaze input through association with different body parts. For example, techniques requiring the alignment of both gaze and hand directions for steering allow directional input to be decoupled from visual attention [37]. Other approaches modify the body part responsible for steering. Regarding the comparison with hand-based steering, results are inconsistent: some studies suggest that gaze-directed steering is superior [35], while others report contrary results [4]. Alternatively, torso-based steering techniques have demonstrated improvements in naturalness, efficiency, and maneuverability [38]. And recent work shows that leaning-based steering, which is also a torso-based steering, can outperform hand-based control during dual-task situations, reducing interference between locomotion and manual interaction and enhancing overall performance and user experience [13]. However, these approaches primarily substitute one input modality for another and do not address the underlying limitation: physical movements used for perception, balance, or task interaction remain statically coupled with the system’s control parameters. Because user intent—whether to explore the environment visually or to initiate locomotion—fluctuates dynamically with context, any static mapping risks recurrent conflicts between natural bodily behavior and system interpretation.

2.3 Adaptive Motion Coupling in Locomotion

In human locomotor control, the coordination dynamics among the eyes, head, and torso are not static but constitute a hierarchically organized system that is dynamically reconfigured depending on the required range of motion. According to Saeb et al., gaze shifts smaller than 20° are accomplished primarily through ocular movements, and the contribution of the head remains relatively small—approximately one third—up to around 30° [32]. Furthermore, Sidenmark et al. reported that the torso contributes substantially to rotational behavior only during large-amplitude gaze shifts exceeding 100° [33]. These findings indicate the existence of a functional decoupling zone in which the torso operates mechanically independently from gaze direction during small-amplitude exploratory behaviors. During high-speed locomotion or turning maneuvers, however, this coordination relationship transitions toward anticipatory coupling. In such contexts, trajectory prediction takes precedence over gaze stabilization, and both the head and torso rotate or tilt toward the forthcoming movement trajectory to lead the body’s heading. As demonstrated by Imai et al., the gaze can substantially precede the heading direction at high angular velocities [15], and the orientation of the head is controlled to align with the gravitational-inertial acceleration (GIA) vector.

In contrast, virtual reality (VR) environments have been shown to reorganize these natural coordination patterns, often resulting in dynamics that diverge from those observed in the real world. Pfeil et al. found that VR users rely more heavily on head movements than on ocular movements when orienting toward targets [28],

while Horsak et al. reported that the mere use of a head-mounted display (HMD) increases mediolateral torso variability and induces posture-coupling patterns that differ from those in physical environments[14]. Speed-dependent coordination also exhibits VR-specific alterations: Giphart et al. demonstrated that low-speed walking in VR destabilizes arm–leg coordination, causing a premature transition to 1:1 coupling compared to natural walking[10]. Brument et al. further showed that when VR navigation methods violate the natural speed–curvature relationship (the 1/3 power law), the magnitude and timing of anticipatory reorientation of the eyes, head, and shoulders differ markedly from those during real-world locomotion[7].

Taken together, real-world locomotion exhibits a hierarchical coordination structure in which the relationship among gaze, head, and torso transitions flexibly and continuously between decoupling and coupling depending on movement amplitude and locomotor speed. In VR, however, these natural coordination structures may be reorganized in ways that diverge from physical behavior, and it remains unclear to what extent real-world coordination dynamics and threshold structures generalize to VR settings[23]. Moreover, VR display characteristics and locomotion interfaces can directly influence users' perception of affordances, potentially altering the interdependence of gaze control, postural regulation, and self-motion perception[12]. Therefore, achieving bio-congruent 3D locomotion in VR requires integrating the speed-dependent coordination principles and natural threshold structures observed in real-world behavior into the underlying VR control mechanisms.

3 Methodology

Our core design philosophy is to mimic the natural decoupling zone found in human physiology. Research indicates that gaze shifts under 20° are purely ocular, while the head contributes significantly up to 30° before the torso engages [32]. Unlike previous methods that rotate the virtual body linearly with the head, AdaptiveFly introduces a non-linear control manifold. We suppress torso rotation for head yaws within the $\pm 30^\circ$ range, creating a stable zone for visual inspection without unintended steering.

Beyond static thresholds, we incorporate the anticipatory mechanism observed in high-velocity locomotion [15]. We formulated a dynamic coupling factor k that increases with the user's angular velocity. As the user turns their head faster, the system predicts a trajectory change and actively rotates the virtual torso into the turn, aligning the view with the Gravito-Inertial Acceleration (GIA) vector.

3.1 Personalized CoP Translation

A primary limitation of static leaning-based interfaces is their “one-size-fits-all” approach. They impose a fixed force threshold (e.g., a 5cm lean) that completely ignores the significant inter-person variability in natural postural sway. This makes the interface feel unresponsive for some users and overly sensitive for others.

To address this, we introduce a personalized calibration mechanism. This system is founded on the first of our bio-congruent principles: the system must first understand the user's unique, natural body dynamics. During a brief calibration phase (approx. 30

seconds), the user is asked to stand still and relaxed on the force platform. We capture the Center of Pressure (CoP) data and model the user's natural sway distribution as a bi-variate Gaussian ($\mathcal{N}_{sway}$).

This statistical model allows us to create a “smart deadzone” tailored to each user. The relationship between this model and the control activation is illustrated in Figure 2 (c). The control function (red curve) is explicitly designed to apply minimal force within the range of the user's natural, unintentional sway (blue curve). Force is only applied when the user's CoP shift (c) significantly exceeds their personalized baseline, for example, beyond two standard deviations ($2\sigma_{cop}$).

The translational force F_g applied to the virtual avatar is calculated using a normalized sigmoid function, as shown in Equation 1. Here, c represents the magnitude of the CoP shift from the calibrated center, and σ_{cop} is the standard deviation of the user's sway measured in the direction of that shift.

$$\frac{F_g}{F_{max}} = \frac{1}{1 + e^{-\alpha(c - \gamma\sigma_{cop})}} \quad (1)$$

Following the visualization in Figure 2 (c), the parameters $\alpha \approx 1.465$ and $\gamma = 3.5$ are used to ensure the force is 10% at $2\sigma_{cop}$ and 50% at $3.5\sigma_{cop}$.

3.2 Context-Adaptive Yaw Control

The second critical limitation of static interfaces is the “Look vs. Steer” conflict. When stationary or moving slowly, users will want to decouple their head rotation from their body's direction to observe the environment. Conversely, when moving quickly, they would want their head and body to be tightly coupled to steer. Static mappings cannot accommodate both, breaking transparency.

Our solution is inspired by human locomotion, which is highly context-dependent. Studies show that during locomotion, a change in direction is tightly coupled with a preceding head rotation, and this coupling becomes much stronger than at rest. AdaptiveFly mirrors this bio-congruence. We propose a Context-Adaptive Yaw control that dynamically adjusts the coupling between head rotation and yaw rate based on the user's current velocity.

The control manifold is illustrated in Figure 2 (b). This mechanism transparently supports two distinct modes:

- **Low-speed exploration:** When velocity v_d is low, the system uses a large deadzone ($\mu_{max} \approx 30^\circ$). This allows the user to freely look around without causing unintended steering.
- **High-speed travel:** When velocity v_d is high, the system reduces the deadzone to a small value ($\mu_{min} \approx 12^\circ$). This creates a tight, responsive coupling, allowing the user's head gaze to act as a precise steering mechanism.

The yaw rate $\dot{\phi}_d$ is calculated based on the absolute value of the user's head yaw angle $|\phi_h|$ using Equation 2:

$$\dot{\phi}_d = \frac{\beta}{1 + e^{-\alpha(|\phi_h| - \mu_v)}} \quad (2)$$

where μ_v is the critical adaptive offset (the deadzone), α controls the response sharpness, and β is the maximum yaw velocity. The adaptive offset is not fixed, but is itself a function of the forward velocity v_d , as shown in Equation 3:

$$\mu_v = \frac{\mu_{min} - \mu_{max}}{1 + e^{k(v_d - v_{th})}} + \mu_{max} \quad (3)$$

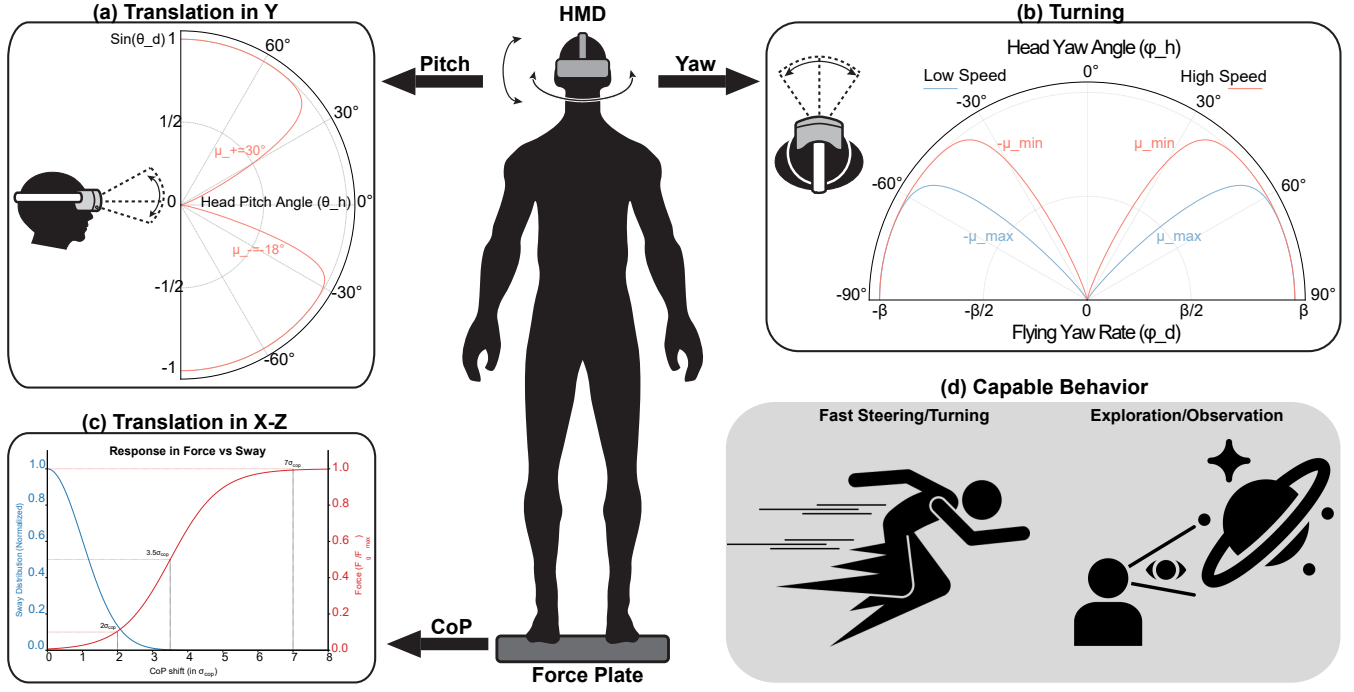


Figure 2: System Overview. The user controls 3D movement via HMD orientation and a force plate. The system features: (a) Asymmetric pitch mapping for vertical translation; (b) Context-Adaptive Yaw control that modulates the Look-and-Steer coupling based on flight velocity; and (c) Personalized CoP calibration to filter unintentional sway. (d) This bio-congruent approach resolves the trade-off between agile steering and stable observation.

Here, μ_{min} and μ_{max} are the minimal and maximal deadzone angles, v_{th} is the velocity threshold, and k controls the smoothness of the transition.

3.3 Asymmetric Pitch Control

Finally, to enhance comfort and naturalness during vertical movement (the 3rd and 4th DoF), we implemented a third bio-congruent mechanism for elevation control, mapped to the user’s head pitch.

AdaptiveFly’s elevation control utilizes an asymmetric mapping. This design is motivated by physiological studies on human gaze, which show that the natural contribution of head movement is smaller for downwards gazeshifts compared to those directed upwards[20, 33].

As shown in Figure 2 (a), the control mapping is designed to be less sensitive for downward head pitches (θ_h) than for upward pitches. This aligns with the natural kinematic constraints of the neck and mitigates uncomfortable or extreme movements, particularly when looking down.

The elevation angle θ_d is computed from the user’s head pitch θ_h using a Sigmoid function (Equation 4), which is applied with two different parameter sets:

$$\sin(\theta_d) = \frac{1}{1 + e^{-\alpha(\theta_h - \mu)}} \quad (4)$$

This function is parameterized with (α_+, μ_+) for positive pitches and (α_-, μ_-) for negative pitches to create the bio-congruent asymmetry.

4 Experiment

To validate the effectiveness of AdaptiveFly, we conducted a user study designed to evaluate its performance, motion quality, and user experience across different flight tasks. We compared AdaptiveFly against traditional joystick control and two ablated mapping embodied flight baselines.

4.1 Participants

We recruited 12 participants (6 female, 6 male). The mean age of the participants was 25.5 years ($SD = 2.62$). The participants possessed varying levels of VR and gaming experience, ranging from zero to over 20 years. Two additional participants withdrew from the study during the first round of experiments due to severe cybersickness; their data were excluded from the analysis. All participants had normal or corrected-to-normal vision, were informed of the experimental procedures and potential risks, and provided written informed consent. The experimental protocol was approved by the local Research Ethics Committee (Approval No. [E2025ALS216-01]).

4.2 Apparatus

The experiment was conducted in an environment equipped with a high-performance PC, using the Unity 6 engine to render the virtual scenes. Participants wore a Meta Quest 3 Head-Mounted Display (HMD) for immersive visual feedback and head motion tracking. We used a force plate to measure changes in the participant’s Center of Pressure (CoP), implemented using the Flexel modular floor system

[36]. In our setup, Flexel provided a 6×6 grid of sensing cells (36 cells in total), each derived from quad-pod load sensors, and the load distribution was sampled at 80 Hz. Additionally, a standard gamepad joystick was used as the baseline condition for abstract control.

4.3 Study Design and Conditions

The study employed a within-subjects design with Locomotion Mode as the independent variable. To rigorously evaluate the contribution of our adaptive mechanism, we compared our method against three baselines:

- **AdaptiveFly (Our Method):** Embodied control utilizing a Context-Adaptive Yaw/Pitch coupling mechanism.
- **CoupledFly (Fully Coupled):** A 1:1 "Gaze-to-Steer" embodied control, serving as an ablation baseline with the adaptive mechanism disabled (coupling strength fixed at $\alpha = 1.0$).
- **StaticFly (Fixed Threshold):** Embodied control using fixed thresholds for rotation and pitch. Steering and elevation are activated via a discrete mechanism: exceeding a 30° head rotation deadzone triggers a constant $60^\circ/s$ turn rate and maximum vertical velocity.
- **Joystick (Baseline):** Control via a standard gamepad, serving as a baseline for abstract control.

In all embodied control modes, horizontal movement was controlled by CoP displacement (body lean) on the force plate.

4.4 Tasks

To comprehensively evaluate the performance of different locomotion conditions across scenarios, we designed two distinct tasks covering navigation and observation:

- **Task 1. Helix Navigation (Precision):** Participants were required to fly along a predefined helical path marked by a series of dots. This task aimed to assess precise path-following capabilities, as shown in Figure 3.
- **Task 2. Orbit Observation (Stability):** Participants were required to orbit around a central target while keeping their gaze fixed on it. This task evaluated the ability to decouple view direction from travel direction (resolving the Look-Steer conflict).

4.5 Procedure

The experimental session began with an explanation of the study's purpose and the signing of informed consent forms, followed by the completion of a baseline cybersickness questionnaire. Next, a personalized CoP calibration was performed: participants stood still on the force plate to calculate their sway distribution and then leaned in four cardinal directions to establish their natural range of motion. To minimize bias, participants were not explicitly informed of the specific locomotion mechanics unless they failed to discover the control rules within two minutes.

Subsequently, they entered a practice scene containing static objects like rings and columns and a flock of dynamic fish-like boid agents based on the model by Reynolds [31], implemented by Lague [21]. Participants practiced by freely chasing the boids or navigating through obstacles until they reported proficiency. Once

familiarized, they completed two tasks under each condition; the order was counterbalanced using a 4×4 Latin Square. The entire procedure is illustrated in Figure 3. Finally, after each condition, participants filled out subjective assessment questionnaires (cybersickness questionnaire first), with mandatory breaks enforced between blocks.

4.6 Data Collection and Metrics

We collected both objective performance metrics and subjective assessment data. For objective metrics, system data was recorded at 60 Hz and pre-filtered. We analyzed Motion Quality via vertical and angular jerk (RMS) to assess smoothness, while Task Performance was evaluated by completion time for Task 1 and gaze offset stability (RMSE) for Task 2. Regarding subjective metrics, we employed a modified Simulator Sickness Questionnaire (SSQ) [18]. To balance sensitivity with efficiency, we selected items covering Nausea, Oculomotor, and Disorientation; this ensured the capture of gastric symptoms induced by continuous locomotion—often omitted by the shorter VRSQ [19]—while avoiding less relevant autonomic symptoms to minimize participant fatigue. Additionally, we measured cognitive load using raw NASA-TLX, Sense of Embodiment (SoE) (Agency, Bodily Sensation, Self-Location), and custom ratings for interaction transparency, flow, and enjoyment.

5 Analysis and Results

We used a within-subjects analysis for all measures. For most objective metrics and subjective ratings, we employed the non-parametric Friedman test to analyze overall differences among conditions. When the Friedman test indicated significant effects, post-hoc pairwise comparisons were conducted using the Wilcoxon signed-rank test with Holm correction. For NASA-TLX overall scores, the data met normality assumptions; therefore, we used a repeated-measures ANOVA and report the corresponding F statistic. The significance level was set at $\alpha = .05$.

5.1 Task Performance and Motion Quality

5.1.1 Task 1: Movement. In the precision navigation task, locomotion mode had a significant effect on both performance and motion quality.

Performance: Significant differences were found in completion time ($\chi^2(3) = 13.70, p = .003$). As shown in Figure 4 (a), CoupledFly was the fastest ($M = 47.38s$), significantly faster than Joystick ($M = 88.67s, p = .029$). AdaptiveFly ($M = 54.19s$) was also faster than Joystick ($p = .061$, marginally significant). Path offset RMSE also showed significant differences ($\chi^2(3) = 8.20, p = .042$). CoupledFly achieved the highest precision ($M = 3.13m$), significantly outperforming AdaptiveFly ($M = 3.87m, p = .009$). We also visualized all trajectories in Figure 5.

Motion Quality: Average angular velocity showed extremely significant differences ($\chi^2(3) = 27.10, p < .0001$). As shown in Figure 4 (b), the angular velocity of CoupledFly ($M = 31.76^\circ/s$) was significantly higher than that of AdaptiveFly ($M = 7.21^\circ/s$) and StaticFly ($M = 8.49^\circ/s$) ($p < .003$). Furthermore, angular Jerk RMS differed significantly ($\chi^2(3) = 21.90, p < .0001$); AdaptiveFly was the lowest, significantly lower than all other conditions ($p < .003$), indicating the smoothest steering control.

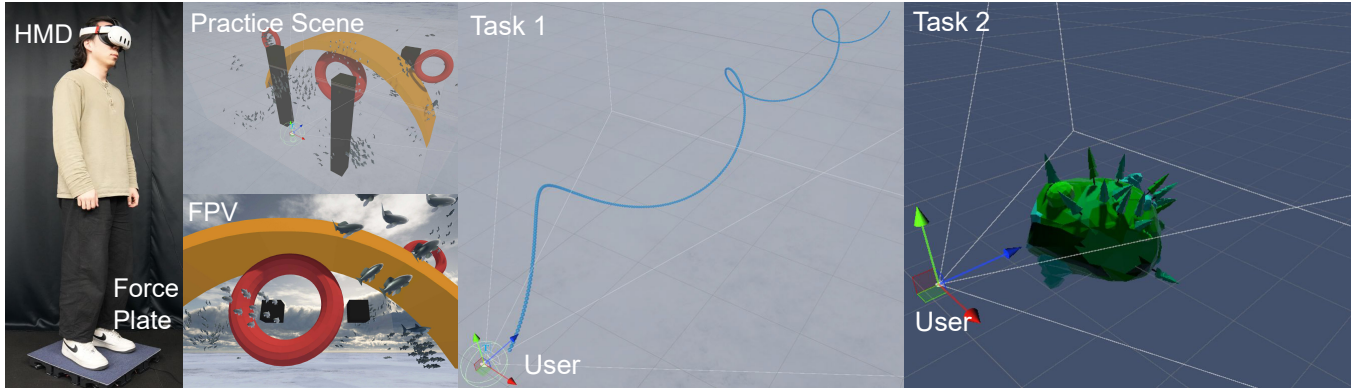


Figure 3: Experiment Setup. Participants wear an HMD and stand on a force plate to calibrate. They then familiarize themselves with each motion mode in the practice scene by freely moving, before proceeding to Task 1 and Task 2.

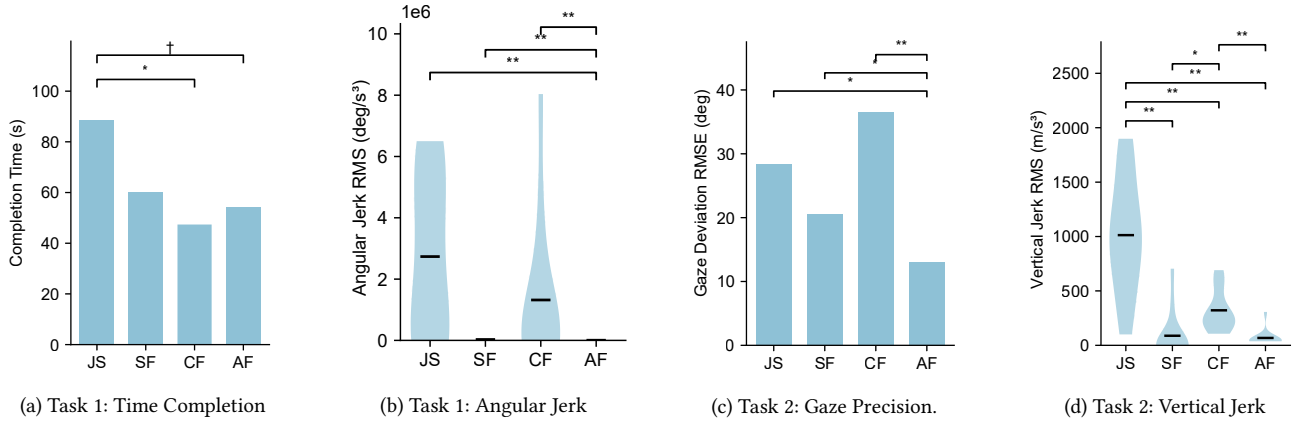


Figure 4: Objective Performance and Motion Quality Metrics. (a) Completion Time for Task 1. (b) Angular Jerk RMS, indicating steering smoothness. (c) Gaze Deviation RMSE for Task 2, measuring observational stability. (d) Vertical Jerk RMS. AdaptiveFly (AF) demonstrates a balance of fast completion times and significantly lower jerk compared to CoupledFly (CF) and Joystick (JS). ($\dagger < .1$, $** p < .05$, $*** p < .01$)

In terms of navigation tasks, AdaptiveFly performed comparably to CoupledFly, with both conditions significantly outperforming the traditional Joystick baseline. Crucially, however, AdaptiveFly demonstrated a distinct advantage over CoupledFly in terms of motion quality, exhibiting significantly lower jerk metrics. This reduction in jerk implies a smoother motion profile that is less likely to induce vestibular conflict. These results suggest that AdaptiveFly offers a locomotion paradigm that is not only faster and more intuitive than abstract joystick control but also objectively superior in user experience, maintaining high performance while mitigating cybersickness.

5.1.2 Task 2: Observation. This task was the critical scenario for testing the resolution of the "Look vs. Steer" conflict.

Performance: Gaze offset RMSE showed highly significant differences ($\chi^2(3) = 17.70, p = .0005$). As shown in Figure 4 (c), AdaptiveFly demonstrated superior performance ($M = 13.10^\circ$), significantly outperforming all other conditions: Joystick ($M = 28.38^\circ, p < .05$), StaticFly ($M = 20.49^\circ, p < .05$), and CoupledFly ($M = 36.56^\circ, p < .003$).

Notably, CoupledFly performed worst in this task. Orbital stability RMSE also differed significantly ($\chi^2(3) = 9.10, p = .028$), with AdaptiveFly ($M = 2.775m$) significantly better than Joystick ($M = 5.226m, p < .015$). We also visualized all trajectories and participants gaze direction in Figure 6.

Motion Quality: Extremely significant differences were found in vertical Jerk RMS ($\chi^2(3) = 29.40, p < .0001$) and vertical acceleration RMS ($\chi^2(3) = 29.40, p < .0001$). As shown in Figure 4 (d), AdaptiveFly achieved the lowest values on both metrics, significantly lower than Joystick and CoupledFly ($p < .01$), indicating the most stable and smooth control in the vertical direction.

Results in Task 2 strongly validate the effectiveness of AdaptiveFly in resolving the Look-Steer conflict. While the CoupledFly suffered from severe unintentional steering during visual inspection, AdaptiveFly successfully allowed users to decouple their gaze for observation without disrupting their flight trajectory. Furthermore, the significantly lower jerk metrics confirm that this adaptive decoupling not only improves task accuracy but also ensures a

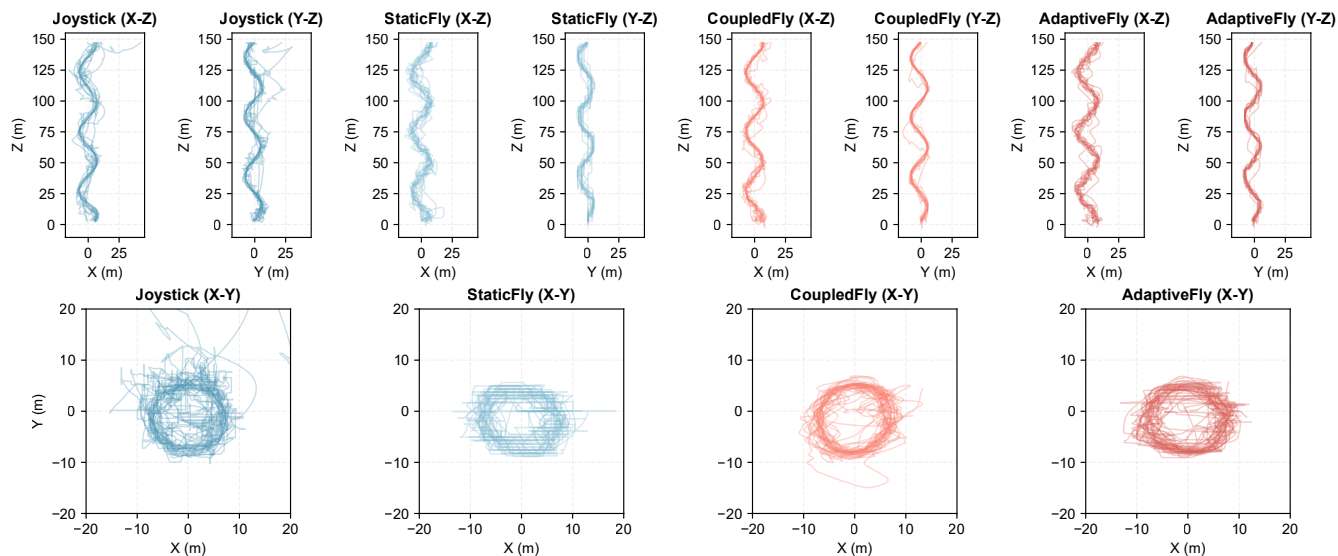


Figure 5: Trajectory Visualizations for Task 1. Top-down (X-Z) and side views (X-Y, Y-Z) of participant paths across four conditions.

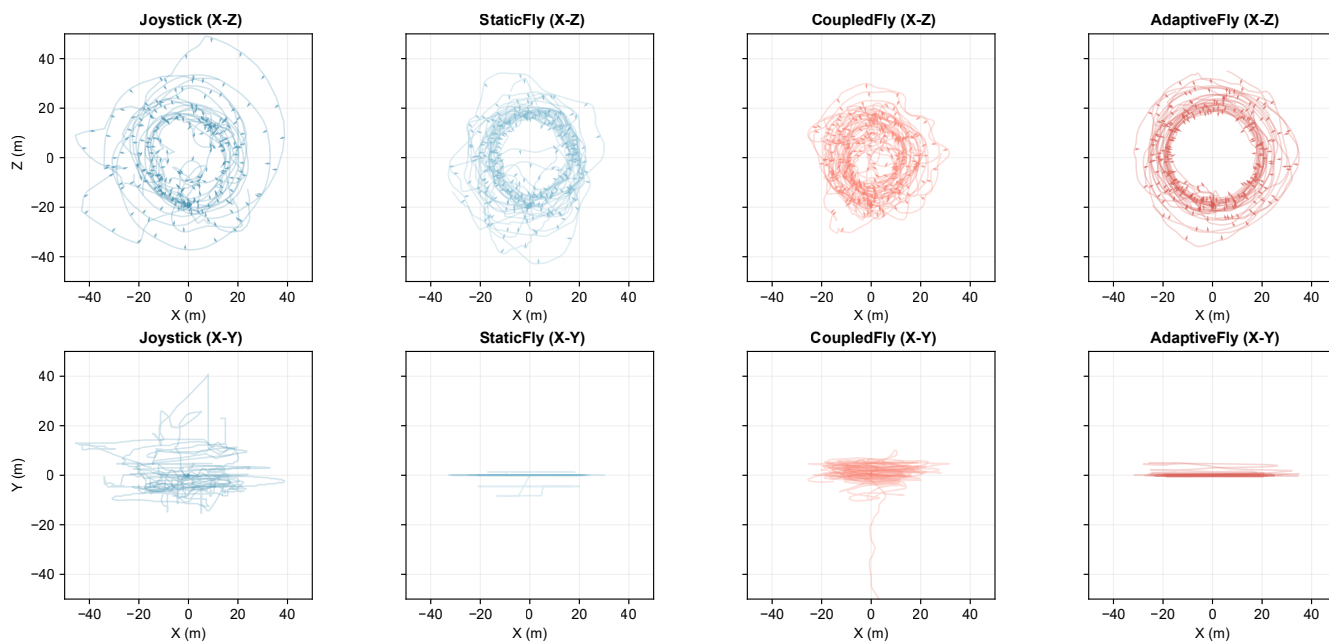


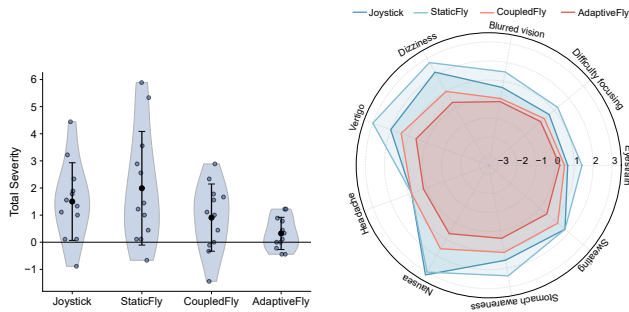
Figure 6: Trajectory and Gaze Visualizations for Task 2. This task required orbiting a central target. Joystick and CoupledFly traces show irregular deviations and corrections. AdaptiveFly enables a consistent circular orbit, confirming that users could decouple their head gaze for observation without disrupting their flight path.

more stable and comfortable hovering experience compared to both abstract and static embodied baselines.

5.2 Subjective Experience

5.2.1 Cybersickness (Modified SSQ). We analyzed the SSQ scores relative to baseline (Delta Scores). As shown in Figure 7 a, for Total Severity, although statistical significance was not reached ($p = .082$), AdaptiveFly yielded the lowest mean score ($M = 0.324$), whereas StaticFly had the highest ($M = 1.991$). Figure 7 b further illustrates that AdaptiveFly outperformed all other conditions across every dimension evaluated. It is worth noting that we observed high standard deviations across all conditions, reflecting the significant individual variability in Cybersickness susceptibility. Despite this inherent variability, AdaptiveFly consistently yielded the lowest Cybersickness scores with minimal variance. This suggests that our approach has a strong tendency of improvement across users with varying degrees of VR sensitivity, effectively mitigating discomfort for a broader demographic rather than just a specific subgroup.

Regarding specific symptoms, significant differences were found in Dizziness ($\chi^2(3) = 10.05, p = .018$), with AdaptiveFly ($M = 0.333$) inducing the least dizziness; post-hoc comparisons showed that AdaptiveFly was marginally lower than Joystick ($M = 2.167, p = .117$) and StaticFly ($M = 2.750, p = .082$). Furthermore, significant differences were observed in Blurred Vision ($\chi^2(3) = 11.50, p = .009$), where StaticFly ($M = 1.500$) caused the most severe symptoms, significantly higher than AdaptiveFly ($M = -0.083, p < .05$).



(a) Rain cloud of overall cybersickness score (b) Modified SSQ Radar Chart

Figure 7: Modified SSQ results. (a) Rain cloud plot showing the overall score distribution. (b) Radar chart illustrating the symptom profile, where AdaptiveFly demonstrates reduced severity in Oculomotor and Disorientation components.

5.2.2 Cognitive Load (Raw NASA-TLX). Overall cognitive load showed significant differences ($F(3, 33) = 4.93, p = .006$). Joystick had the highest cognitive load ($M = 4.278$), while CoupledFly was the lowest ($M = 3.722$). A marginally significant difference existed between Joystick and CoupledFly ($p = .063$). AdaptiveFly ($M = 3.944$) remained at a lower level.

5.2.3 Sense of Embodiment (SoE). Among the three dimensions of embodiment, *Self-Localization* showed significant differences ($\chi^2(3) = 9.72, p = .021$). Joystick had the lowest sense of self-localization ($M = 4.722$), significantly lower than CoupledFly ($M =$

$5.500, p < .05$). AdaptiveFly ($M = 5.556$) achieved the highest score. No significant differences were found for Agency and Bodily Sensation across conditions.

5.3 Qualitative Feedback

Participant interview feedback provided valuable context for the quantitative results and revealed the pros and cons of different control modes.

AdaptiveFly (Balance & Focus): Many participants felt AdaptiveFly achieved a good balance between comfort and control. P4 felt AdaptiveFly was "just right." In the observation task (Task 2), AdaptiveFly was particularly favored (P1, P2, P10). P5 noted that with AdaptiveFly, they could best "immerse in observation" and "focus on the task itself," whereas other modes required "figuring out how to control the viewpoint." P10 considered AdaptiveFly "the most comfortable" and "probably the best overall."

CoupledFly (Speed vs. Error): CoupledFly was appreciated for its high responsiveness and speed, especially in tasks requiring quick reactions. P1 thought it "moved very smoothly, suitable for Task 1." However, its high sensitivity was also a major drawback. P4 felt it was "too sensitive." P10 noted it was "too easy to misoperate; turning the head makes you dizzy easily." This drawback was particularly evident in the observation task Task2, with P2 explicitly stating that CoupledFly was "the hardest to use" in Task2.

StaticFly (Lag & Discomfort): StaticFly was generally criticized, with many participants reporting severe discomfort and control difficulties (P1, P9, P11 all stated StaticFly was the dizziest). P3 felt "frame drops and a sense of lag, easy to lose control of direction." This "disconnection between body movement and will" (P5) was considered the primary cause of discomfort.

Joystick (High Load & Low Immersion): As a baseline, Joystick was perceived as difficult to control with high cognitive load. P2 and P10 rated Joystick extremely poorly. P5 felt Joystick made them "focus on control rather than observation." P12 also mentioned that, apart from Joystick, the other modes "were all very immersive."

6 Implementation to Physical Platform

While the previous study validated AdaptiveFly within a controlled virtual environment, assessing the ecological validity of the system requires deployment on physical hardware. In this section, we describe a proof-of-concept implementation where the control scheme was transposed onto a commercial Vertical Take-Off and Landing (VTOL) quadcopter. This deployment aimed to verify the technical feasibility of the architecture and identify critical hardware requirements for embodied drone flight.

6.1 System Integration

We utilized a DJI Tello Ryze quadcopter as the physical platform. The interface architecture remained identical to the VR implementation (see Figure 8). Regarding control mapping, the velocity commands generated by the virtual drone model were scaled down by a constant factor and transmitted to the physical drone via Wi-Fi; to ensure safety, the velocity vector magnitude was capped at 10 cm/s. The yaw rate was mapped in a 1:1 ratio, ensuring the drone's heading strictly followed the user's head rotation. Finally, for visual feedback, the video stream (720p, 30 FPS) from the drone's fixed

front-facing camera was retrieved and rendered directly into the user's HMD to provide first-person view (FPV) feedback.

6.2 Field Evaluation

We conducted a qualitative evaluation using a "Fly Through Waypoints" (FTW) task. Users were asked to navigate the drone through physical hoops (diameter ≈ 70 cm) spaced five meters apart and positioned 1-2 meters above the ground (Figure 8).

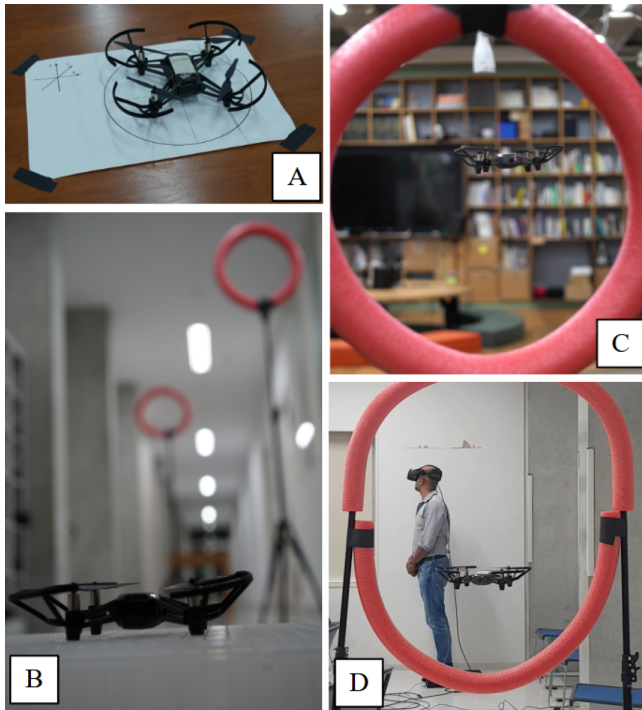


Figure 8: Real-world Deployment. (A) The DJI Tello Ryze drone used for physical validation. (B) The First-Person View (FPV) streamed to the user. (C, D) A user performing the "Fly Through Waypoints" task, demonstrating the transferability of the AdaptiveFly control interface from VR to physical robots.

6.3 Observations and Challenges

Technically, the system successfully translated body-based inputs into drone actuation with minimal latency. However, piloting the physical drone proved significantly more cognitively demanding than the simulation. This pilot study revealed two critical hardware constraints that currently hinder the full potential of embodied flight in real-world settings:

The consumer-grade camera provided a low-resolution, monocular video stream. Unlike the 3D rendered environment in VR, this 2D feed lacked binocular depth cues, making distance estimation challenging. Furthermore, the restricted Field of View (FOV $\approx 82.6^\circ$) meant users lost visual contact with the hoops up to 50 cm before traversal. This forced users to rely on optical flow estimation

and mental trajectory extrapolation rather than continuous visual feedback, significantly increasing cognitive load.

A more fundamental issue arose from the lack of a mechanical gimbal. In our embodied scheme, users look up (pitch up) to command ascent. However, since the camera is fixed to the drone's chassis, the visual horizon does not tilt up with the user's head; it only tilts if the drone physically pitches, which is often decoupled from vertical velocity in quadcopters. This created a sensory conflict: the user's proprioception signaled "looking up," but the visual feedback remained level. This mismatch disrupted the Sense of Embodiment and spatial orientation, particularly during pure elevation changes. Users had to actively suppress the expectation that the camera would follow their head pitch, requiring careful motion planning.

This transposition confirms that AdaptiveFly is technically agnostic to the target platform (virtual or physical). However, the qualitative results strongly suggest that for embodied control to be effective in the real world, the drone hardware must match the sensory fidelity of VR. Specifically, a stereoscopic camera (for depth) and a synchronized 2-axis gimbal (to align camera pitch with head pitch) are prerequisites to resolving the sensory conflicts we observed.

7 Discussion

The primary goal of this study was to determine if a bio-congruent approach could resolve the tension between high performance and the sense of embodiment in VR locomotion. We proposed AdaptiveFly, a 4-DoF embodied flight paradigm, to test this hypothesis. Our results achieved both high task performance and a strong self-localization, while significantly improving user comfort.

Our findings highlight the inherent limitations of static mappings. CoupledFly, which uses a fixed Gaze-to-Steer mapping, performed as expected in the precision task (Task 1) where looking and steering are naturally aligned. However, this advantage came at a significant cost to stability. In the observation task (Task 2), CoupledFly performed the worst, showing much higher gaze offset than all other conditions. This confirms that when a user wants to look around without turning, a fully coupled system creates severe interference.

AdaptiveFly, by comparison, proved flexible enough to handle these conflicting task demands. In precision task (Task 1), it performed nearly as well as CoupledFly and significantly better than the Joystick. More importantly, it outperformed all baselines in the observation task. This features directly from our core contribution: Context-Adaptive Yaw control. When flying fast, the system tightens the coupling for precision; at low speeds, it loosens the control, allowing users to look around freely. As noted by Participant 5, this makes the control feel "transparent," letting users focus on the task rather than the interface.

Looking at the motion quality metrics helps explain why AdaptiveFly felt better to use. A key finding was that CoupledFly resulted in extremely high angular velocity and jerk—often double that of other modes. This suggests that users were constantly fighting the system, making rapid corrections to counteract unintended steering caused by their head movements. In contrast, AdaptiveFly produced the smoothest motion profile with the lowest jerk. The

adaptive deadzone effectively filtered out involuntary head movements, making steering intentional and precise. We believe this smoothness is a major factor in reducing cybersickness. Indeed, AdaptiveFly yielded the lowest symptom scores, particularly for dizziness and blurred vision. Conversely, the poor performance of StaticFly (which used fixed thresholds) led to the most severe discomfort. Participant feedback about "lag" and "stuttering" in the StaticFly condition suggests that violating natural motion expectations, even with an embodied interface can be worse than using a simple abstract controller.

Our physical drone deployment (Section 6) served as a "reality check" for the ecological validity of our method. While the control logic transferred successfully from VR to the real robot, the user experience highlighted the critical role of sensory fidelity. We observed a "Pitch-Gaze" conflict: users would pitch their heads up to fly up, but the drone's fixed camera remained level. This mismatch broke the sense of embodiment and increased cognitive load. This reinforces our argument for bio-congruence: embodied interfaces work best when visual feedback matches vestibular and proprioceptive expectations. Future physical implementations will need hardware that matches human anatomy, such as a synchronized gimbal.

Despite the significant advantages demonstrated by AdaptiveFly, this study has limitations. First, our sample size ($N = 12$) was relatively small; future research could validate these findings with a larger and more diverse sample. Second, while our physical deployment demonstrated technical feasibility, the current lack of stereoscopic vision and mechanical gimbals on the test drone limited the evaluation of complex maneuvers. Future work should evaluate AdaptiveFly on robotic platforms equipped with synchronized 2-axis gimbals and stereo cameras to fully replicate the sensory fidelity of the VR condition. Furthermore, AdaptiveFly's current adaptive mechanism is primarily based on velocity as context information. Future work could explore integrating more biological signals, such as eye-tracking data, to more accurately infer user observation intent.

8 Conclusion

Traditional static mapping methods in VR locomotion often struggle to balance high performance with natural embodiment. To address this, we propose AdaptiveFly, which introduces bio-congruent adaptive mechanisms to reconcile these demands. By integrating Context-Adaptive Yaw and Personalized CoP calibration, the system dynamically adjusts control strategies based on the interaction context, effectively resolving the persistent 'Look vs. Steer' conflict.

Through a comprehensive user study, we demonstrated that AdaptiveFly not only maintains high performance in precision navigation and dynamic efficiency tasks but also significantly outperforms existing methods in tasks requiring independent observation and stability. Simultaneously, AdaptiveFly provides smoother motion quality, enhances the user's Self-Location, and reduces Cybersickness. Furthermore, our implementation on a commercial quadcopter confirms that this bio-congruent paradigm is platform-agnostic, capable of controlling not just virtual avatars but also physical robots, provided that sensory feedback aligns with user expectations. Our research highlights the importance of context

adaptation in achieving transparent control and offers new directions for the future design of embodied 3D user interfaces.

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